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ADAPTIVE REVERSALS IN ACID TOLERANCE IN COPEPODS FROM LAKES RECOVERING FROM HISTORICAL STRESS

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Abstract. Anthropogenic habitat disturbance can often lead to rapid evolution of environmental tolerances in taxa that are able to withstand the stressor. What we do not understand, however, is how species respond when the stressor no longer exists, especially across landscapes and over a considerable length of time. Once anthropogenic disturbance is removed and if there is an ecological trade-off associated with local adaptation to such an historical stressor, then evolutionary theory would predict evolutionary reversals. On the Boreal Shield, tens of thousands of lakes acidified as a result of SO_2 emissions, but many of these lakes are undergoing chemical recovery as a consequence of reduced emissions. We investigated the adaptive consequences of disturbance and recovery to zooplankton living in these lakes by asking (1) if contemporary evolution of acid tolerance had arisen among *Leptodiaptomus minutus* copepod populations in multiple circum-neutral lakes with and without historical acidification, (2) if *L. minutus* populations were adaptively responding to reversals in selection in historically acidified lakes that had recovered to pH 6.0 for at least 6–8 years, and (3) if there was a fitness trade-off for *L. minutus* individuals with high acid tolerance at circum-neutral pH.

L. minutus populations had higher acid tolerances in circum-neutral lakes with a history of acidification than in local and distant lakes that were never acidified. However, copepods in circum-neutral acid-recovering lakes were less acid-tolerant than were copepods in lakes with longer recovery time. This adaptive reversal in acid tolerance of *L. minutus* populations following lake recovery was supported by the results of a laboratory experiment that indicated a fitness trade-off in copepods with high acid tolerances at circum-neutral pH. These responses appear to have a genetic basis and suggest that *L. minutus* is highly adaptive to changes in environmental conditions. Therefore, restoration managers should focus on removing environmental stressors, and adaptable species will be able to reverse evolutionary responses to environmental disturbance in the years following recovery.

Key words: acidification history; adaptive reversal; anthropogenic disturbance; Boreal Shield lakes; copepods; ecosystem recovery; environmental tolerance; fitness trade-off; rapid evolution; local adaptation; pH; zooplankton.

Introduction

Anthropogenic impacts can cause rapid evolutionary change, which in turn can alter genetic and adaptive properties of populations over a scale of decades (Palumbi 2001). There are numerous examples of adaptively diverged populations associated with anthropogenic stressors (Reznik and Ghalambor 2001) that have potential ecological consequences for populations, communities, and ecosystems (Hairston et al. 2005). Restoration managers need to consider contemporary evolution in degraded and recovering habitats because application of artificially augmented gene flow from other habitats to enhance biological recovery can disrupt the ability of existing populations to adaptively track environmental changes (Stockwell et al. 2003).

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A limited number of studies on recovering ecosystems have shown evolutionary reversals following decreased pollution: loss of metal resistance in oligochaetes after environmental remediation (Levinton et al. 2003) and a decline in the melanic frequency of moths in post-industrial England (Cook et al. 2005). However, it is unclear how populations locally adapted to a particular stressor will respond once the disturbance is removed because a genetically fixed trait resulting from local adaptation to a historical stressor may continue to be expressed even after an environmental change causes it to lose its advantage (Stearns 1994). Also, reversals in the direction of selection may not necessarily return a population to its earlier state (Grant and Grant 2002). Although certain species may be able to cope adaptively, population declines associated with environmental change are most often associated with local extinction (Ricciardi and Rasmussen 1999).

Acid deposition and its interactions with climate change, land use, and stratospheric ozone depletion have had huge regional effects, particularly on boreal lakes (Schindler 1998) and oceans (Orr et al. 2005). Lake acidification caused losses of hundreds of thousands of fish and invertebrate populations (Minns et al. 1990). One exception was a copepod species, *Leptodiaptomus minutus* Lilljeborg, one of the most common crustacean zooplankton on the Boreal Shield (Carter et al. 1980; see Plate 1). This species can be found in shield lakes across a broad pH range (pH 4–7) and tends to dominate zooplankton communities in acid-damaged lakes (Sprules 1975). There is evidence from an experimentally acidified lake in Wisconsin, USA, that some populations of *L. minutus* may evolve increased tolerance to low pH conditions (Fischer et al. 2001). However, we do not know if this is a response that can be generalized to the thousands of lakes that have been damaged by regional acidification.

As lakes damaged by acid precipitation chemically recover with declining SO_2 emissions across North America and Europe (Stoddard et al. 1999, Jeffries et al. 2000), we do not know if there will be evolutionary responses to reversals in selection forces, i.e., will organisms lose their tolerance to low pH? Copepod populations may have undergone rapid local adaptation following atmospheric acidification of lakes on the Canadian Shield, as they did in Little Rock Lake, Wisconsin (Fischer et al. 2001), but they may now be facing reverse selection as lakes recover to circum-neutral pH, especially if there is a cost associated with increased acid tolerance. Although biological recovery is one of the ultimate goals for SO_2 emission control programs (Meteorological Service of Canada 2004), knowledge about genetic aspects of acidification recovery and its potential significance is scarce.

The objectives of this study were to examine patterns of acid tolerance in *L. minutus* in lakes that differ in acidification history and explore evolutionary reversals in recovering lakes. First, we tested whether or not rapid evolution of acid tolerance had arisen among *L. minutus* populations in multiple circum-neutral lakes with and without historical acidification in a boreal landscape by using field transplant experiments. We included replicate lakes from both local and regional spatial scales to determine if dispersal among local lakes had homogenized acid tolerance adaptation across the local landscape. Theoretical work suggests that dispersal among habitats promotes plasticity over local adaptation, by mixing gene pools and broadening environmental tolerances of locally specialized genotypes (Sultan and Spencer 2002), which would result in a uniform response from different populations among lakes.

Next, we asked if there have been evolutionary reversals in acid tolerance after several years of lake-water recovery to pH 6.0. We used a second field transplant experiment to test if there are differences in acid tolerances among *L. minutus* populations between circum-neutral acid-recovering lakes and anthropogenic and naturally acidic lakes. The circum-neutral lakes used in this experiment had varying numbers of years since recovery to pH 6.0, allowing us to investigate the relationship between acid tolerance and years since recovery. If reversals in the direction of evolution of acid tolerance had occurred, we expected to see a decreasing acid tolerance associated with longer times since recovery to pH 6.0.

Finally, we investigated if there is a fitness trade-off associated with adaptation of high acid tolerance as acidified lakes recover to pH 6.0. We conducted a laboratory experiment that compared survival in copepods with high acid tolerance to those with low acid tolerance, at both circum-neutral and low pH.

Methods

Study site

Killarney Provincial Park, Ontario, Canada (46°01′ N, 81°24′ W) contains >600 lakes with pH ranging from 4.3 to 7.6 (Snucins et al. 2001). This wide range in pH is the result of both natural pH gradients and anthropogenically induced pH gradients associated with SO_2 emissions from nearby metal smelters during the mid-1900s and the long range transport of acid deposition. The circum-neutral buffered lakes contain limestone and sandstone in their watersheds (Debicki 1982), and therefore, high concentrations of base cations that are able to neutralize acid deposition. Bog lakes have low pH ($\leq$5.3) resulting from organic acids from associated wetlands. Anthropogenically acidified lakes were historically circum-neutral lakes (Dixit et al. 2002) not associated with limestone and sandstone and therefore, were highly susceptible to acidification (Sprules 1975). Many lakes in Killarney Park are starting to recover to circum-neutral pH (pH $\geq$ 6.0) with declining SO_2 emissions, but recovery rates are dependent on watershed properties, hydrology, and probably the degree of acidification (Snucins et al. 2001, Keller et al. 2003). We chose to use pH 6.0 as a threshold for recovery of *L. minutus* because it is a widely used measure for biological recovery of acid-sensitive species in the acidification recovery literature (Keller et al. 2002). In Killarney Provincial Park, therefore, there are a mosaic of lakes with different acid-stress histories that we classified as: (1) circum-neutral lakes that did not acidify with regional acid deposition; (2) circum-neutral, acid-recovering lakes that had a pH $\leq$ 5.2 during the 1970s but have had a pH > 6.0 for at least six years, (3) acid-damaged lakes that were acidic during the 1970s because of regional acid deposition and continue to have a pH $\leq$ 5.2; and (4) bog lakes with pH $\leq$ 5.3 (Sprules 1975, Dixit et al. 2002).

Experiment 1: field transplant with circum-neutral lakes

We hypothesized that greater acid tolerance had arisen in copepod populations from circum-neutral lakes with a history of acidification than in nearby and regionally distant (>200 km, near Dorset, Ontario) unacidified lakes. To test this, we conducted a factorial-

TABLE 1. Selected physicochemical characteristics of Canadian study lakes included in field transplant and laboratory experiments.

Lake history	Lake name	Lat. (N)	Long. (W)	Max. depth (m)	Volume $\times 10^4$ (m^3)	Watershed area (ha)	pH	DOC (mg/L)	TP (µg/L)	1% UVB attenuation depth (m)
Regional	Bigwind†	45°03′	78°04′	32.0	1180	507	6.8	4.0	6.5	0.4
Un-acidified	Blue Chalk†	45°11′	78°56′	23.0	447	158	6.9	2.4	3.1	0.7
(Dorset)	Red Chalk‡	45°11′	78°56′	38.0	811	589	6.7	4.6	3.2	0.4
Local	Helen	46°06′	81°33′	41.2	1692	883	6.9	4.1	7.0	0.4
Buffered	Ishmael	46°06′	81°35′	19.8	820	1243	7.0	4.1	5.8	0.4
(Killarney)	Low	46°06’	81°33′	28.4	483	1000	7.6	3.0	3.4	0.6
	Teardrop§	46°02′	81°24′	16.6	32	13	6.8	1.1	8.0	1.5
Local	Bell (C1, H1)	46°08′	81°11′	26.8	2827	8596	6.3	5.5	5.4	0.3
Acid-recovering	Carlyle (C1)	46°05′	81°14′	14.6	891	1059	6.3	4.1	5.2	0.4
(Killarney)	David|| (C2)	46°08′	81°17′	24.4	2857	1915	5.8	1.9	3.0	0.9
	George (H1, C2)	46°01′	81°24′	36.6	3086	5717	6.2	2.4	5.0	0.7
	Johnnie|| (H1, C1, C2)	46°05′	81°14′	33.5	3418	12 952	6.1	3.6	5.0	0.5
Local	Little Superior¶	46°04′	81°19′	33.5	179	35	4.2	0.2	4.0	5.4
Acid-damaged	OSA||	46°03′	81°24′	39.7	3341	879	4.7	0.3	2.4	4.1
(Killarney)	Proulx¶	46°04′	81°19′	28.7	119	42	4.4	0.2	3.6	5.4
Local	Heaven††	46°04′	81°17′	17.8	9	14	4.6	3.1	<6.0	0.6
Bog	Lake of Woods#	46°06′	81°12′	6.0	24	67	5.3	3.3	<8.0	0.5
(Killarney)	Turbid¶	46°06′	81°11′	9.4	69	1136	5.2	3.2	16.0	0.5

Notes: Lake-water pH was measured at the time of experiments; 2004 midsummer dissolved organic carbon (DOC) and total phosphorus (TP) data were measured by authors for all lakes except where indicated. In parentheses after acid-recovering lake names, letters represent water source treatment (C, Carlyle water; H, home water), and numbers represent experiment (experiment 1 vs. 2). Physical characteristics of study lakes are from Reid et al. (1987) for Blue Chalk and Red Chalk lakes, from Girard et al. (*in press*) for Bigwind Lake, and from Snucins and Gunn (1998) for the Killarney lakes.

† 2004 midsummer chemistry data (Ontario Ministry of the Environment, *unpublished data*).

‡ 2003 midsummer chemistry data (Ontario Ministry of the Environment, *unpublished data*).

§ 2001 midsummer DOC and TP (Ontario Ministry of the Environment, *unpublished data*).

|| 2004 midsummer pH data (Ontario Ministry of the Environment, *unpublished data*).

¶ 2000 midsummer DOC and TP data (Holt and Yan 2003).

1997 winter chemistry data (Snucins and Gunn 1998).

†† 2003 winter DOC (Ontario Ministry of the Environment, *unpublished data*) and 1997 winter TP (Snucins and Gunn 1998).

design field transplant experiment in circum-neutral Carlyle Lake, Killarney Park, Ontario (see Plate 1). We manipulated the following factors: zooplankton acidification history (three levels: buffered, acid-recovering, and un-acidified, regionally distant Dorset), water source (two levels: lake water from Carlyle Lake and water from each of the source lakes from which the zooplankton were collected ["home lake water"]), and pH treatment (three levels: circum-neutral pH 6.2–7.6, 5.4, and 4.7). This experimental design was not completely balanced: $n = 3$ buffered Killarney lakes, $n = 4$ acid-recovering Killarney lakes, and $n = 3$ un-acidified and regionally distant lakes near Dorset, Ontario. These lakes have a current and historical presence of the copepod *L. minutus* (Sprules 1975, Keller et al. 2002). The physicochemical properties of these habitats are described in Table 1. All of the lakes had long-term records of circum-neutral lake-water pH $\approx$ 6.0, except for the acid-recovering lakes, which had been circum-neutral for at least six years previously: Bell Lake (since 1996), George Lake (since 1998), Johnnie Lake (since 1998), and Carlyle Lake (since 1998) (Keller et al. 2003).

Zooplankton from each of the 10 study lakes were stocked at 2× ambient lake densities in 66 closed, translucent 10-L polyethylene containers. A mixed zooplankton assemblage was stocked into these containers to minimize logistical complications, and was not expected to influence direct effects of acid addition on *L. minutus* response since indirect community effects require longer than a week (Webster et al. 1992). Zooplankton were collected with a 15 cm diameter closing plankton net (80-µm mesh) at a 1.5-m depth interval just below 1% UVB attenuation in each source lake. One-percent UVB-attenuation depths (Morris et al. 1995, Table 1) were calculated from dissolved organic carbon (DOC) concentrations for each of the study lakes to ensure that (1) zooplankton were collected from a sufficient depth interval as to account for vertical migration that might result from UVB avoidance, and (2) containers were incubated at sufficient depth to exclude UVB effects in Carlyle Lake. One zooplankton haul from each lake was preserved (5% sugar-formalin) to estimate initial zooplankton densities stocked into containers. Containers were suspended from floating frames in a 10 m deep basin in Carlyle Lake and incubated at 1.5–2.0 m depths at 20°C (YSI 95 O_2/temperature meter; YSI, Yellow Springs, Ohio, USA) for the seven-day period from 13 August to 20 August 2004.

While 33 of the containers were filled with water from Carlyle Lake, the remaining 33 containers were filled with home lake water. Water was collected from the top 5 m of the water column using a battery-operated pump in Carlyle Lake and an integrated tube sampler (5 m long; 2.5-cm internal diameter) in each of the other study lakes. We filtered the water through an 80-μm mesh to remove crustacean zooplankton, but allowed rotifers, phytoplankton, and bacteria to pass through into the containers. Incubation of *L. minutus* in both Carlyle Lake water and home lake water enabled us to differentiate between the possible effects of algal food quality and quantity and potential effects from local adaptation to home-water conditions.

Three acid treatments (circum-neutral pH 6.2–7.6, 5.4, and 4.7) were imposed using 1 mol/L sulfuric acid for each acidification history and water source treatment combination. The pH in each container was measured using a WTW inoLab 740 pH meter (WTW, Woburn, Massachusetts, USA). Two replicates of the buffered home-water treatments (Ishmael and Low lakes) and one replicate of the acid-recovering lake history treatment in Carlyle water (George Lake) were accidentally not acidified to pH 4.7. Otherwise, pH treatments were all within ± 0.2 units of the target pH. Therefore, although there were 66 containers in the experiment, data from 63 containers were statistically analyzed.

At the end of the experiment, 9 L of water from each container was filtered through 80-μm mesh to collect zooplankton that were subsequently preserved in 5% sugar-formalin. The remaining 1 L of water from each of the containers was 80-μm filtered to remove zooplankton, and then a 0.4-L subsample was analyzed for pH (WTW inoLab 740 pH meter). An estimate of total chlorophyll *a* was obtained by collecting phytoplankton from a subsample of 0.3 L on a Fisherbrand 42.5 mm diameter, 1.2-μm G4 glass fiber filter (Fisher Scientific Canada, Nepean, Ontario, Canada). A measure of edible chlorophyll *a* was obtained by first filtering a 0.3-L subsample through a 30-μm mesh and then collecting the phytoplankton from the filtrate on a.1.2-μm glass fiber filter since copepods graze on highly edible algae <30–35 μm (Cyr and Curtis 1999).

Adult *L. minutus* and fifth-stage calanoid copepodites were counted from initial zooplankton samples (N_i) and from samples taken from each container at the end of the experiment (N_f) under a Leica MZ16 dissecting microscope (Leica Microsystems [Canada], Richmond Hill, Ontario, Canada). Final abundance (N_f) was sometimes greater than the initial abundance (N_i) due to maturation of nauplii and early-stage copepodites. Only copepods that had been alive or recently alive at the time of formalin preservation were counted. Individuals that exhibited signs of decay or incomplete body structure were not counted.

Experiment 2: field transplant with acidified lakes

Our second field transplant experiment compared acid tolerance of *L. minutus* populations from circum-neutral acid-recovering lakes with anthropogenic acid-damaged and naturally acidic bog lakes in Killarney Park (Table 1). If *L. minutus* had responded to a reversal in selection in acid-recovering lakes, then we expected to find a lower acid tolerance in acid-recovering lakes than in acid-damaged or bog lakes. To test this hypothesis, we used a 3×4 factorial design manipulating zooplankton acidification history (three levels: from acid-recovering, acid-damaged, and bog lakes) and pH treatment (three levels: circum-neutral $\geq$6.0, 5.4, 4.7, and 3.8). Epilimnetic zooplankton from three lakes in each of the acidification history categories were collected and stocked into 36 closed, translucent 10-L polyethylene containers on 18 June 2005. Because of low abundances of *L. minutus* in Bell and Carlyle lakes at the time of the experiment in 2005, circum-neutral acid-recovering lakes were comprised of George, Johnnie, and David (pH of 5.6 as of 2005) lakes. Zooplankton were incubated in only Carlyle Lake water for this experiment.

Methods for zooplankton collection and stocking paralleled the approach described previously for experiment 1. Containers were suspended from wooden frames and incubated at 1.5–2.0 m depths at 24°C in acid-recovering Carlyle Lake as a common garden experiment for seven days. Four acid treatments within each of the "acidification history" treatments were set based on prior titrations with 1 mol/L sulfuric acid and were maintained within ± 0.2 units of the target pH. As with the first transplant experiment, an initial zooplankton sample from each lake and a final zooplankton sample from each container were preserved in 5% sugar-formalin.

Experiment 3: laboratory assay for chronic pH effects

Finally, we investigated if there is a trade-off associated with high acid tolerance in *L. minutus* at circum-neutral pH. A 32-day two-factor laboratory experiment was conducted for *L. minutus* individuals collected from two side-by-side Killarney lakes with distinct acidification histories and lake-water pH: circum-neutral, buffered Teardrop Lake and anthropogenic acid-damaged OSA Lake (Table 1). Copepods in each lake-type were subjected to two pH levels: neutral (pH 6.7 $\pm$ 0.3) and acidic (pH 4.7 $\pm$ 0.2). Zooplankton were collected from each lake using an 80-μm plankton net, placed in 2-L Nalgene containers in coolers, and transported to the laboratory at Queen's University, Kingston, Ontario. Females without eggs and males (20 adults per replicate) were placed in 100 mL of 0.2-μm-filtered Carlyle Lake water for each replicate ($n = 3$). Algal food was a 1:1 mix of *Chlamydomonas reinhardtii* and *Cryptomonas erosa* (University of Toronto Culture Collection of Algae and Cyanobacteria, Department of Botany, University of Toronto, Toronto, Ontario, Canada) and concentrations ranged from 3 to 6×10^3 cells/mL. Previous feeding experiments had indicated that survival was highest at this food concentration under neutral pH conditions (P. Lam, *unpublished data*). Water and algae were refreshed, pH was adjusted using 1 mol/L sulfuric acid, and nauplii were removed every 3 to 5 days. Copepod and algal cultures were incubated at

20°C under 16:8 h of light : dark photoperiod in a Percival model 1-36LLVL incubator (Percival Scientific, Perry, Iowa, USA). Surviving adults were counted on days 5, 13, 22, and 32 of the experiment under Leica MZ16 dissecting microscopes.

Statistical analyses

For each of experiments 1 and 2, initial (N_i) and final (N_f) *L. minutus* copepod abundances were $\ln(x + 1)$-transformed to normalize data distributions and homogenize variance. These data were analyzed with factorial-design (lake history × water source × pH) analyses of covariance (ANCOVAs) with N_i as a covariate and N_f as a dependent response variable. A covariate (N_i) was used to control for differences in initial stocking densities among lakes rather than standardization by proportions (N_f/N_i) because proportions can cause changes in the covariance and correlation structure of the data matrix that can result in different statistical outcomes compared to raw abundances (Jackson 1997). For experiment 1, assumptions of normality were met, but equal variance was violated. Although the effect of violating these ANCOVA assumptions is a greater risk of accepting a false null hypothesis (Type I error) (Rheinheimer and Penfield 2001), we were able to reject H_0 for the interaction of interest (lake history × pH: see *Results*). For experiment 2, all assumptions were met except for homogeneity of regression slopes. There was a significant effect of lake history on the covariate N_i because initial stocking densities (N_i) were higher in bog (27.2 ± 0.5 *L. minutus*/L; mean ± SE) and acid-damaged lakes (9.9 ± 0.05 *L. minutus*/L) than in acid-recovering lakes (3.4 ± 0.13 *L. minutus*/L) (factorial ANOVA on N_i; lake history main effect, $F = 10.49$, df = 2, 24, $P < 0.01$; Tukey's hsd tests, $P < 0.01$). Violation of homogeneity of regression slopes in an ANCOVA can increase the probability of accepting a false H_0. We conducted a factorial ANOVA on $\ln(N_f + 1)$ without N_i as a covariate and found that the ANCOVA results for experiment 2 were supported by an analysis that was not based on this assumption.

To test if final chlorophyll *a* concentrations and conductivity measurements varied among treatments in experiments 1 and 2, we used factorial ANOVAs. For experiment 1, both normality and equal variance assumptions were violated for total chlorophyll *a,* despite log-transformation, but variances were equal for log-transformed edible chlorophyll *a*. All ANOVA assumptions were met for log-transformed total and edible chlorophyll *a* concentrations in experiment 2.

To investigate the relationship between copepod acid tolerance and time since recovery among acid-recovering lakes, copepod survival ($\ln(N_f/N_i)$) at pH 4.7 from acid-damaged and acid-recovering lake history categories in experiments 1 and 2 was linearly regressed on length of time since lake recovery to circum-neutral conditions.

TABLE 2. Results of ANCOVAs from field transplant experiments 1 and 2 for *Leptodiaptomus minutus* final abundance (N_f) with initial abundance (N_i) as a covariate.

Experiment and factor	df	F	P
1) Circum-neutral lakes			
Covariate (N_i)	1	74.27	**<0.01**
Lake history (LH)	2	5.56	**<0.01**
Water source (WS)	1	2.11	0.15
pH	2	63.66	**<0.01**
LH × WS	2	0.70	0.50
LH × pH	4	4.55	**<0.01**
WS × pH	2	0.83	0.44
LH × WS × pH	4	0.60	0.67
Error	46		
2) Acidified lakes with different histories			
Covariate (N_i)	1	15.76	**<0.01**
Lake history (LH)	2	0.67	0.52
pH	3	34.82	**<0.01**
LH × pH	6	0.43	0.85
Error	23		

Note: Boldface P values indicate significant main and interaction effects ($P < 0.05$).

Regression assumptions of independence, normality, and equal variance were met.

For experiment 3, $\log(x + 1)$-transformed number of alive adult copepods were analyzed with a multivariate repeated-measures ANOVA (RMA) to examine both among- and within-treatment effects. These results were presented despite the fact that most normality and equality of variance assumptions were violated because RMA is robust when sample sizes are equal. Sphericity assumptions were met.

All ANCOVAs and ANOVAs were followed by Tukey's hsd post hoc comparisons. Statistical analyses were conducted with Statistica 6.0 software (Statsoft, Tulsa, Oklahoma, USA).

RESULTS

Experiment 1: field transplant with circum-neutral lakes

Leptodiaptomus minutus final abundance varied both among lake-acidification history categories and pH treatments (Table 2). Copepod abundance was lowest in the pH 4.7 treatments (ANCOVA; Tukey's hsd tests, $P < 0.01$). The effect of pH treatment on copepod final abundance, however, was dependent on lake history, as indicated by a significant lake history × pH interaction (Table 2, Fig. 1). Within the acid-recovering lake history category, *L. minutus* final abundance was not lower at pH 4.7 than at pH 5.4 or circum-neutral pH (Fig. 1). However, there were significant declines in *L. minutus* density at pH 4.7 within circum-neutral buffered and regionally un-acidified Dorset lake history categories compared with pH 5.4 and circum-neutral pH treatments (ANCOVA; Tukey's hsd test, $P < 0.01$; Table 2, Fig. 1). At pH 4.7, *L. minutus* individuals had greater final abundance in the acid-recovering lake history treatment than in the circum-neutral buffered or the regionally un-acidified (Dorset) lake history treatment

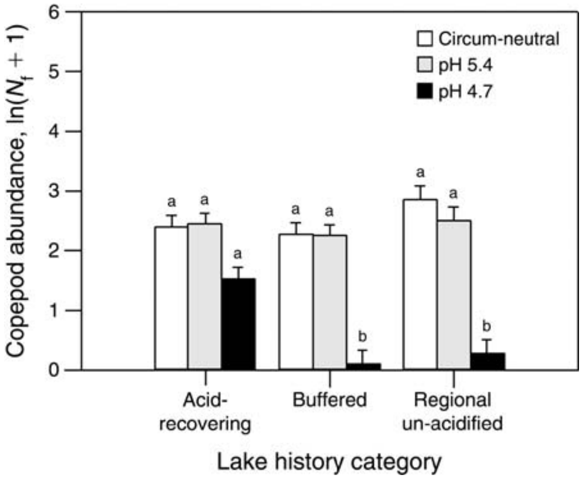

FIG. 1. Final abundance (N_f) of *Leptodiaptomus minutus* copepods (measured as no./L, but log-transformed; adjusted least-squares mean + SE) from each circum-neutral lake history category (local acid-recovering vs. local buffered vs. regional un-acidified Dorset lakes) at each pH treatment: circum-neutral, 5.4, and 4.7. Water source treatments were averaged because we detected no significant differences between them. Lowercase letters indicate pH 4.7 treatments where population abundance was significantly lower than in pH 5.4 and circum-neutral treatments within lake-history categories (ANCOVA; Tukey's hsd tests, $P < 0.01$) after controlling for variation in initial abundances among lakes as a covariate.

(ANCOVA; Tukey's hsd test, $P < 0.01$; Table 2, Fig. 1) across water-source treatments. Overall, no differences in copepod final abundance were detected among home and Carlyle water treatments, as indicated by an insignificant water source main effect, and insignificant water source × lake history and water source × pH interactions (Table 2).

We detected differences in concentrations of log-transformed total (factorial ANOVA, total pH effect, df = 2, $P = 0.03$) and log-transformed edible (factorial ANOVA, edible pH effect, df = 2, $P < 0.01$) chlorophyll *a* among pH treatments. Total and edible chlorophyll *a* concentrations were higher at pH 4.7 (total group mean, 2.68 ± 0.218 μg/L; edible group mean, 2.25 ± 0.12 μg/L) than the pH 6.2 (total group mean, 2.13 ± 0.16 μg/L; edible group mean, 1.60 ± 0.11 μg/L) and 5.4 treatments (total group mean, 2.32 ± 0.13 μg/L; edible group mean, 1.88 ± 0.08 μg/L; all Tukey tests, $P < 0.05$). Carlyle Lake water treatments had lower total chlorophyll *a* concentrations (2.0 ± 0.06 μg/L) than home-water treatments (2.8 ± 0.2 μg/L; factorial ANOVA, water-source effect, df = 1, $P < 0.01$). However, no differences in edible chlorophyll *a* were detected among water source treatments.

Experiment 2: field transplant with acidified lakes

Acid tolerance of *L. minutus* from acid-recovering lakes was similar to that of individuals from acidic lakes; i.e., *L. minutus* abundance did not differ among lake history categories at pH treatments 6.1, 5.4, and 4.7, as indicated by an insignificant lake history main effect and an insignificant pH × lake history interaction (Table 2, Fig. 2). There were significant declines in *L. minutus* density at pH 3.8 within all lake history categories, compared with pH 6.2, 5.4, and 4.7 (ANCOVA; Tukey's hsd tests, $P < 0.01$; Table 2, Fig. 2). There were no significant differences in total and edible chlorophyll *a* concentrations within lake-history categories among or within pH treatments.

Experiments 1 and 2: loss of acid tolerance with time

A loss of acid tolerance in copepods from circum-neutral acid-recovering lakes was supported by a negative linear relationship between years since recovery to pH 6.0 (0–8 years) and copepod survival ($\ln(N_f + 1) - \ln(N_i + 1)$) at low pH among acid-recovering and acid-damaged lakes ($P < 0.01$, adjusted $R^2 = 0.58$). Acid-recovering lakes with longer times since recovery to pH 6.0 had lower *L. minutus* survival at pH 4.7 than more recently recovered lakes ($F_{1,11} = 17.61$, $P < 0.01$; Fig. 3). Although not included in the regression of time since recovery to pH 6.0, copepod survival from bog and un-acidified buffered and Dorset lakes at pH 4.7 were plotted for reference for how far adaptive reversal in acid tolerance has proceeded in these circum-neutral acid-recovering lakes (Fig. 3).

Experiment 3: laboratory experiment for chronic pH effects

Our laboratory experiment revealed that there may be fitness (or ecological) trade-offs associated with in-

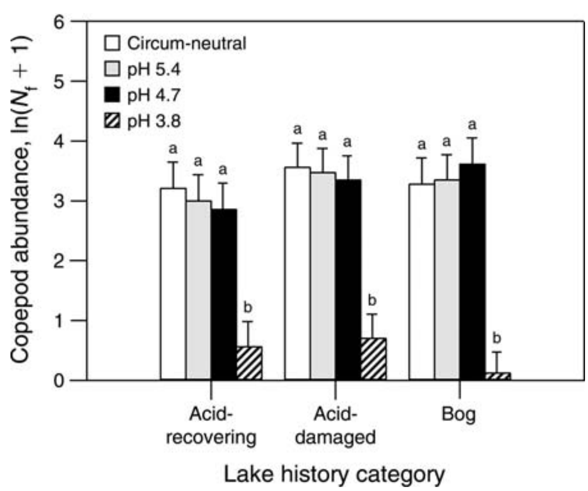

FIG. 2. Final abundance (N_f) of *Leptodiaptomus minutus* copepods (no./L, log-transformed; adjusted least-squares mean + SE) from each acidified lake history category (circum-neutral acid-recovering vs. anthropogenic acid-damaged vs. naturally acidic bog lakes) in Carlyle water at each pH treatment: circum-neutral, 5.4, 4.7, and 3.8. Lowercase letters indicate pH 3.8 treatments where population abundance was significantly lower than other pH treatments within lake-history categories (ANCOVA; Tukey's hsd tests, $P < 0.01$) after controlling for variation in initial abundances among lakes as a covariate.

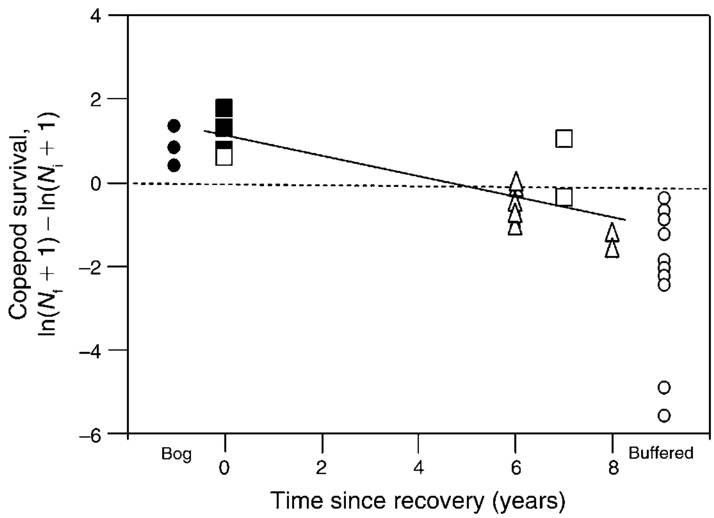

FIG. 3. Time since recovery from acidification in acid-damaged (solid squares, experiment 2) and circum-neutral acid-recovering lakes (open squares, experiment 2; open triangles, experiment 1) were linearly regressed on survival of *Leptodiaptomus minutus* copepods (log-transformed final abundance [N_f] minus initial abundance [N_i]) at pH 4.7. Whereas values at or above zero indicate treatments where there has been no change or an increase in population density, values below zero indicate a decrease in population density from initial concentrations. Although not included as part of the regression, copepod survival from bog (solid circles, experiment 2) and buffered (open circles; un-acidified Killarney and Dorset lakes, experiment 1) sources were included for reference. *L. minutus* survival from Carlyle and home source water treatments at pH 4.7 were included for acid-recovering and buffered lakes from experiment 1, but were not labeled because we did not detect any significant differences among source water treatments in experiment 1.

creased acid tolerance in *L. minutus*. *L. minutus* survival was dependent on both the population source and pH, but populations responded to pH differently through time (Table 3, Fig. 4). Consistent with results from experiment 1, the number of individuals from the circum-neutral buffered lake (Teardrop) significantly declined at acidic pH treatment 4.7 by day 5 (Tukey tests, $P < 0.05$ for Teardrop at pH 4.7 for days 5, 13, 22, and 32), but abundance was maintained at circum-neutral pH 6.8. As expected, *L. minutus* from acid-damaged OSA Lake did not significantly decline in abundance at pH 4.7 ($P > 0.18$, Tukey tests). However, when cultured in lake water at pH 6.8, OSA Lake copepods declined in abundance over time with significantly lower numbers detected by day 32 ($P < 0.01$, Tukey tests). At day 32, abundance of Teardrop Lake copepods at pH 6.8 and OSA Lake copepods at pH 4.7 were similar. However, OSA Lake copepod abundance at pH 6.8 was lower than Teardrop at pH 6.8, and OSA copepod abundance was higher at pH 4.7 than Teardrop at pH 4.7 ($P < 0.05$, Tukey tests).

DISCUSSION

Our work found evidence of rapid local adaptation of copepod populations in multiple lakes that experienced historical regional acidification. Increased acid tolerance in *L. minutus* populations was found not only in acidic lakes, but also to varying degrees in recovering lakes that have been circum-neutral for six to eight years. These results generalize those of Fischer et al. (2001) who also found increased acid-tolerance of *L. minutus*, but from a single, experimentally acidified lake in northern Wisconsin, USA. Our study demonstrates that *L. minutus* populations are able to respond to anthropogenic lake acidification with adaptive solutions that parallel acid tolerances of copepod populations found in naturally acidic bog lakes. Bog lakes can have similarly

TABLE 3. Repeated-measures ANOVA results for experiment 3.

Factor	df	F	P
Lake population (buffered vs. acidic)	1	11.87	**<0.01**
pH (6.8 vs. 4.7)	1	28.772	**<0.01**
Lake population × pH	1	64.95	**<0.01**
Error	8		
Time × lake population	3	4.44	**0.01**
Time × pH	3	4.29	**0.01**
Time × lake population × pH	3	18.96	**<0.01**
Error	24		

Note: Boldface P values are statistically significant ($P < 0.05$).

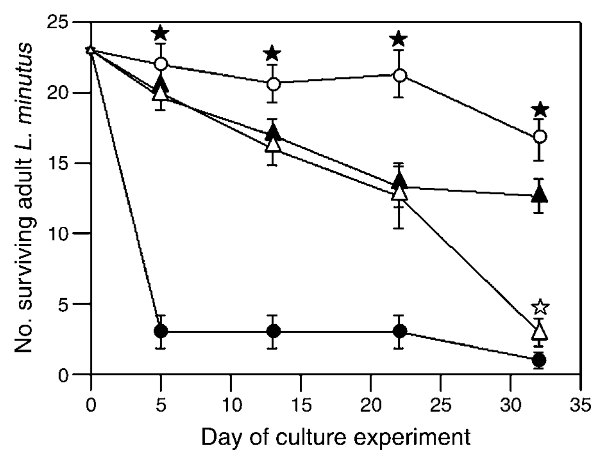

FIG. 4. *Leptodiaptomus minutus* adult abundance (mean ± SE) from a circum-neutral, buffered lake (Teardrop; pH = 6.5; circles) and an acidic, acid-damaged lake (OSA; pH = 5.0; triangles) when cultured in the laboratory for 32 days at pH 6.8 (open symbols) and pH 4.7 (closed symbols). Solid stars indicate significant differences in abundance for *L. minutus* from Teardrop Lake at pH 6.8 vs. 4.7. The open star indicates a significant difference in abundance for *L. minutus* from OSA Lake at pH 6.8 vs. 4.7 on day 32 (repeated-measures ANOVA; Tukey's hsd test, $P < 0.05$).

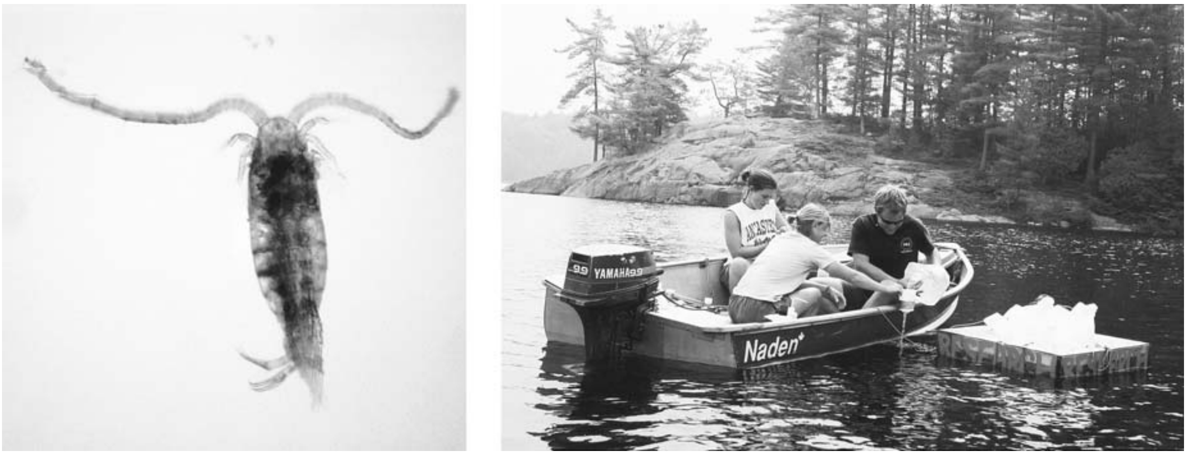

PLATE 1. (Left) The study organism, *Leptodiaptomus minutus* (male). Photo credit: Jessica Forrest. (Right) Collecting results from the field transplant experiment on Carlyle Lake. Photo credit: Justin Shead.

low pH as anthropogenic-acidified lakes, but contrast in other properties such as DOC. While studies have shown that stress resistance can rapidly evolve across multiple similar habitats (Jasieniuk and Maxwell 1994, Räsänen et al. 2003, Baker et al. 2004, Sarnelle and Wilson 2005) and multiple species (Klerks and Weis 1987, Jasieniuk and Maxwell 1994, Rosenheim et al. 1996, Cook et al. 2005) in a region, none to our knowledge have coupled this with a comparison of natural environmental gradients.

Rapid evolution of stress tolerance may partially explain the wide distribution of *L. minutus* throughout lakes on the Canadian Shield (Carter 1971) despite the acidification of thousands of water bodies from industrial sources of SO_2 (Schindler 1998). While many populations decline or are extirpated in the face of anthropogenic stressors, there is increasing evidence that some populations thrive owing to contemporary evolution of stress tolerance (Palumbi 2001). The proliferation of locally adapted genotypes may enable compensatory dynamics analogous to species replacements that maintain ecosystem functioning during disturbance (Velland and Geber 2005).

Although copepods from our circum-neutral lakes with a history of acidification showed increased acid tolerance compared to those from lakes with no history of acidification, they appear to be losing their acid tolerance over time. *L. minutus* in lakes with a greater length of time since recovery to pH 6.0 had lower tolerances to acidic conditions than those in lakes more recently recovered to circum-neutral pH. This apparent reversal in acid tolerance is likely the result of a fitness trade-off associated with high acid tolerance under circum-neutral water conditions, as demonstrated by the decline in the number of acid-adapted copepods that survived in neutral lake water during our laboratory study. This fitness trade-off occurred in the absence of potentially negative biotic interactions that would occur under natural conditions, suggesting that physiological constraints may be driving the reversal in acid tolerance. Although few studies have been conducted on recovering systems, adaptive reversals have been detected in two other studies (loss of metal tolerance in estuarine oligochaetes, Levinton et al. [2003]; loss of industrial melanism in moths, Cook et al. [2005]).

Other studies have also shown that adaptive responses can carry a cost. For example, grasses adapted to high metal concentrations are poor competitors with metal intolerant genotypes in uncontaminated soil (McNair 1981), and the decline of melanic moths following legislation designed to improve air quality has been attributed to selective predation by birds (Cook et al. 2005). Similar to a physiological trade-off for acid-adapted copepods in neutral lake water, rapid loss of metal tolerance in oligochaetes following restoration of an estuary was likely due to a trade-off between cadmium adaptation and mortality rate in non-contaminated sediment (Levinton et al. 2003).

While our laboratory experiment showed lower fitness of acid-tolerant copepod individuals at circum-neutral pH after the duration of a month, tolerance to low pH was maintained by *L. minutus* copepods even after 6–8 years of lake recovery to pH 6.0. *L. minutus* in Little Rock Lake, Wisconsin, were still acid-tolerant after two years of recovery to circum-neutral pH (Fischer et al. 2001). There was no evidence of intra-annual pH variation that could have delayed the loss of copepod acid tolerances in our acid-recovering study lakes. These lakes have low flushing rates (large volumes relative to catchment areas, Table 1) and biweekly measurements of epilimnetic pH in George Lake in 2004 confirmed low variability in pH (coefficient of variation = 0.03) that did not drop below 6.0 (A. Kalyniuk, *unpublished data*). Although acid-tolerant individuals from OSA Lake were able to survive long enough to reproduce nauplii in circum-neutral lake water (A. Derry, *unpublished data*), a reduced fitness at circum-neutral pH may result in a gradual decline in acid tolerance at the population level only after several years of recovery to pH 6.0.

The negative relationship between acid tolerance and time since recovery could result from either reversals in selection as lakes recover to pH 6.0 or the severity of acidification that was experienced (if more acidic lakes take longer to recover). The local geology of boreal shield watersheds has influenced both the severity and duration of lake-water acidification (Snucins et al. 2001, Keller et al. 2003). For example, David Lake is in a poorly buffered watershed that contains orthoquartzite bedrock (Snucins et al. 2001), severely acidified to pH 4.6 in 1972 (Sprules 1975), and is only just beginning to recover to circum-neutral pH conditions (pH 5.6 in 2005). As a result, *L. minutus* copepods from David Lake had high acid tolerances that were comparable to individuals from acid-damaged lakes with pH < 5.0 (Fig. 3). *L. minutus* populations collected from the other four circum-neutral acid-recovering lakes have catchments with more easily weathered bedrock formations (Snucins et al. 2001) and all of these lakes historically acidified to approximately the same extent (Bell, pH 5.2; Carlyle, pH 5.0; George, pH 4.9; and Johnnie, pH 5.1 in 1972; Sprules 1975). In spite of similarly severe historical acidification among these four lakes, *L. minutus* survival from these populations indicated a loss of acid tolerance over 6–8 years of recovered lake-water pH (Fig. 3). Although it is difficult to discern the effects of time since recovery to pH 6.0 and severity of acidification on acid tolerance of copepods from acid-recovering lakes, we showed that copepod populations that had become locally adapted to historical acidification became less acid tolerant within a decade of exposure to recovered circum-neutral pH.

The differences in acid tolerance that we observed among *L. minutus* from lakes of different acidification history were most likely a direct effect of acid toxicity. Although we detected differences in food resource quantity among treatments, food limitation was not likely a factor in copepod survival since the most abundant chlorophyll *a* concentrations were detected at pH 4.7 where copepod abundance was lowest.

The differences that we detected in copepod acid tolerance among lakes with different acidification histories likely resulted from heritable genetic adaptation rather than a phenotypic response to environmental conditions. First, the loss of acid tolerance in acid-recovering lakes is unlikely a result of reinvasion by nonresistant genotypes because our mitochondrial DNA genetic work (A. Derry, *unpublished data*) indicates that *L. minutus* is spatially dispersal limited. Phenotypic (nongenetic) differences in traits among populations may occur if there are differences in phenotypic plasticity or maternal effects. Phenotypic plasticity, the production of multiple phenotypes from a single genotype under different environmental conditions, is unlikely because we did not detect a uniformly broad pH tolerance from different copepod populations among lake history categories. However, it is possible that there have been changes in genetic $\times$ environment interactions or selection for evolution of changes in plasticity (Pigliucci 2005).

Maternal effects, phenotypic variation in offspring that is a consequence of the mother's phenotype rather than the offspring's genetic makeup (Kawecki and Ebert 2004), could not solely account for the differences in acid tolerance that we detected. Both maternal effects and local adaptation could transfer acid tolerance to the next generation, but maternal effects are generally minimized after rearing for two or three generations under common environmental conditions (Kawecki and Ebert 2004). With the exception of David Lake, all acid-recovering lakes in our experiments had circum-neutral pH conditions for at least six years (Keller et al. 2003). *L. minutus* are obligately sexual organisms that most often recruit new individuals into populations by means of subitaneous eggs that hatch within several days in the water column. Under favorable conditions, a single female can produce hundreds of subitaneous eggs over her lifetime (one to several months; Williamson and Reid 2001). We estimate that multiple, sequential generations (at least 50–75 generations in five years assuming an estimate of 10–12 generations per year) under circum-neutral conditions had passed, which would have minimized the possibility of maternal effects, especially with high population densities.

Maternal effects could contribute to our observed acid tolerance in lakes with a history of acidification if there was high recruitment of individuals, whose parents directly experienced acid conditions, from dormant resting eggs that accumulate in sediments within lakes through time (Williamson and Reid 2001). However, it takes 6–8 years to accumulate 1–2 cm of sediment in Killarney lakes (Dixit et al. 2002), such that older eggs containing acid-adapted genotypes are buried below 2 cm or greater of sediment. Hatching cues penetrate only the first 1–2 cm of surface sediments and are confined to shallow near-shore areas (Brendonck and De Meester 2003), restricting potential emergence to a small portion of the lake bottom surface area. Given that there are likely low recruitment rates of acid-adapted genotypes from sediment egg banks, and that several generations of *L. minutus* must have occurred in the water column to reach pelagic densities, we hypothesize that acid tolerance in *L. minutus* is a heritable trait that has been passed from one generation to the next in acid-recovering lakes that have been circum-neutral for under a decade.

Environmental stressors, both natural (Schluter 2000) and anthropogenic (Palumbi 2001), can promote adaptive divergence of populations. High genetic diversity can insure a broad spectrum of genotypes with different environmental optima that can buffer against ecosystem change in the event of changed environmental conditions in a manner analogous to compensatory species with similar ecological functions (Vellend and Geber

2005). For example, recent work on a coastal community highlighted the role of genotypic diversity within one cosmopolitan eelgrass species in augmenting community recovery following an unprecedented heat wave (Reusch et al. 2005). Genetic diversity may be analogous to species and functional group diversity in providing ecosystem resilience to disturbance. Our preliminary work on *L. minutus* in boreal shield lakes suggests that mitochondrial DNA haplotypic diversity is not reduced in anthropogenically acidified lakes compared to buffered lakes (A. Derry, *unpublished data*), suggesting that there may have been high capacity for adaptation, even after selection for acid tolerance. In a landscape where many thousands of lakes in southeastern Canada are beginning to recover to pH 6.0 as a result of SO_2 emission controls (Jeffries et al. 2000) and where biological recovery is lagging behind chemical recovery (Keller et al. 2002), the ability of *L. minutus* to maintain genetic diversity by adapting to pH change may assist the reestablishment of acid-sensitive aquatic communities and higher trophic levels by sustaining an intermediate energy link during both the historical acidification and subsequent recovery of boreal lakes. This adaptive link may be important for a long time since the latest National Acid Rain Assessment (Meteorological Service of Canada 2004) has reported that hundreds of thousands of lakes in eastern Canada will remain acidified at current levels of acid deposition.

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